SmartRoute Pro — Statistical Analytics Report

Subject: Statistical Data Analysis (SDA) • Parametric, Non-Parametric & Regression Analytics

1. Univariate Parametric & Non-Parametric Metrics

Sample variance (s^2) and sample standard deviation (s) are computed using Bessel's correction (ddof = 1) to ensure unbiased estimators: s^2 = (1 / (n - 1)) sum (x_i - mean)^2.

Statistical Parameter	Consignment Distance	Road Network Edge	Dijkstra Route
Sample Count (n)	8	9	2
Sample Mean (x■)	9.62 km	7.34 km	7.95 km
Median (Q2)	7.90 km	6.20 km	7.95 km
Sample Variance (s², ddof=1)	24.736	13.663	126.405
Sample Std Dev (s)	±4.97 km	±3.70 km	±11.24 km
Interquartile Range (IQR)	7.08 km	2.50 km	7.95 km
Min / Max Range	4.5 / 18.2 km	4.3 / 16.5 km	0.0 / 15.9 km

2. Bivariate OLS Linear Regression: Road Transit Time

Regression Parameter	Estimated Value
Sample Size (n pairs)	9
Pearson Correlation (r)	**0.9837**
Coefficient of Determination (R^2)	**0.9676**
OLS Fitted Equation	**y = 1.646x + 3.912**
Two-Tailed p-value	0.0000e+00
Statistical Significance ($\alpha = 0.05$)	**Not Significant**

CAVEAT: *Correlation does not imply causation. Statistical correlation measures linear association strength between distance and transit duration, but unobserved confounding variables (e.g. traffic congestion, delivery delays, weather) may also influence delivery times.*